

Wipro Capstone Project 5

Day 1: Design UI layout and gather system data using system calls.

```
farhanlinux@farhanlinux: ~/Desktop/2141006001/LOS/Day01
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS$ cd Day01
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day01$ g++ day1.cpp
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day01$ ./a.out
total ram:7920MB
total ram:7GB
total ram:4848MB
total ram:4GB
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day01$
```

Day 2: Display process list with CPU and memory usage.

```
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS$ g++ --std=c++17 activeprocess.cpp -o activeprocess
c1plus: fatal error: activeprocess.cpp: No such file or directory
compilation terminated.
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS$ cd Day02
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day02$ g++ --std=c++17 activeprocess.cpp -o activeprocess
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day02$ ./activeprocess
active processes :
ID: 1 | Name: (systemd)
ID: 2 | Name: (kthreadd)
ID: 3 | Name: (pool_workqueue_release)
ID: 4 | Name: (kworker/R-rcu_g)
ID: 5 | Name: (kworker/R-rcu_p)
ID: 6 | Name: (kworker/R-slab_)
ID: 7 | Name: (kworker/R-netns)
ID: 9 | Name: (kworker/0:0H-events_highpri)
ID: 12 | Name: (kworker/R-mm_pe)
ID: 13 | Name: (rcu_tasks_kthread)
ID: 14 | Name: (rcu_tasks_rude_kthread)
ID: 15 | Name: (rcu_tasks_trace_kthread)
ID: 16 | Name: (ksoftirqd/0)
ID: 17 | Name: (rcu_preempt)
ID: 18 | Name: (migration/0)
ID: 19 | Name: (idle_inject/0)
ID: 20 | Name: (cpuhp/0)
ID: 21 | Name: (cpuhp/1)
ID: 22 | Name: (idle_inject/1)
ID: 23 | Name: (migration/1)
```

```
farhanlinux@farhanlinux: ~/Desktop/2141006001/LOS/Day02
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day01$ cd .
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day01$ cd ..
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS$ cd Day02
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day02$ g++ day2.cpp
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day02$ ./a.out
User Time:38363
Nice Time:156
System Time:13110
Idle Time:2567208
Iowait Time:1024
Irq Time:0
Softirq Time:73
Steal Time:0
Guest Time:0
Guest Nice Time:0
Toatal CPU Time : 2619934
99.9769%
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day02$
```

Day 3: Implement process sorting by CPU and memory usage.

```
farhanlinux@farhanlinux: ~/Desktop/2141006001/LOS/Day03
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day02$ cd ..
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS$ cd Day03
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day03$ g++ sorting.cpp
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day03$ ./a.out
PID      CPU%    Memory (kB)    Name
1985     6.59674 0              (gnome-shell)
9483     4.34165 0              (nautilus)
10240    1.31579 0              (a.out)
10030    0.922155 0              (gnome-terminal-)
10234    0.185874 0              (tracker-extract)
8331     0.145146 0              (gjs)
5439     0.102741 0              (soffice.bin)
1        0.0797946 0              (systemd)
1728     0.0732797 0              (spice-vdagentd)
8382     0.0712863 0              (kworker/u12:4-events_power_efficient)
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day03$
```

Day 4: Add functionality to kill processes.

```
farhanlinux@farhanlinux: ~/Desktop/2141006001/LOS/Day04
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day03$ cd ..
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS$ cd Day04
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day04$ g++ killprocess.c
cc1plus: fatal error: killprocess.c: No such file or directory
compilation terminated.
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day04$ g++ killprocess.cpp
farhanlinux@farhanlinux:~/Desktop/2141006001/LOS/Day04$ ./a.out
ti
PID      CPU%    Memory (kB)    Name
1985     6.36179 0              (gnome-shell)
10267    4       0              (a.out)
9483     3.35674 0              (nautilus)
10030    0.687439      0              (gnome-terminal-)
10261    0.190476      0              (tracker-extract)
8331     0.11401 0              (gjs)
5439     0.0960682    0              (soffice.bin)
1        0.0762 0              (systemd)
1728     0.0703297    0              (spice-vdagentd)
8102     0.0632216    0              (kworker/u12:1-events_power_efficient)
Enter PID to Kill :
10267
Killed
```

Day 5: Implement real-time update feature to refresh data every few seconds.

```
farhanlinux@farhanlinux: ~/Desktop/2141006001/LOS/Day5
PID      CPU%    Memory (kB)    Name
1985     5.46702 0              (gnome-shell)
9483     1.64133 0              (nautilus)
10030    0.358546     0              (gnome-terminal-)
10462    0.153139     0              (tracker-extract)
10331    0.13453 0              (gjs)
5439     0.074914     0              (soffice.bin)
10418    0.0746547    0              (kworker/u12:2-events_unbound)
1        0.0652674    0              (systemd)
793      0.0613073    0              (systemd-oomd)
1728     0.0608331    0              (spice-vdagentd)
Enter PID to Kill :
9483
Process 9483 terminated successfully

Press 'q' to quit or Enter to refresh : █
```